

Platform Master Architecture & Team Implementation Blueprint

End-to-End System Specifications, Personas, API Contracts & Cross-Functional Deliverables

TARGET AUDIENCE	STATUS	DATE	REPOSITORY
ALL Engineering Teams	Architecture Approved (100% Tested)	September 2026	Jack-Of-Aces/dev-Ledgr

01. Executive Vision & Platform Mechanics

DevLedgr is built on a non-negotiable premise: “A ledger, not another resume clone.” Traditional tech hiring relies on unverified LinkedIn claims and leetcode memorization. DevLedgr flips the paradigm into an **immutable engineering proof-of-work protocol**.

Developers solve authentic, production-grade distributed systems and operational failure challenges, execute their implementations against live mock container fleets, pass automated CI test harnesses, and receive a **cryptographically stamped certificate valid for 365 days**.

The 3 Immutable Core Invariants of DevLedgr			SYSTEM LAWS
INVARIANT	PRINCIPLE	SYSTEM ENFORCEMENT	
Mutation Safety Rule	Write actions (POST/PUT/DELETE) must <i>never</i> fall back to silent mock success.	Strict error throws in <code>httpClient.ts</code> prevent phantom ledger updates.	
Dual-Layer RBAC	Access control is validated both at Edge compute and client UI rendering.	<code>proxy.ts</code> (Edge middleware) + <code>AuthGuard.tsx</code> component.	
Cryptographic Validity	Proofs are keyed to Git commit SHAs with guaranteed 365-day tamperproof validity.	SHA-256 ledger stamps with public verification URL at <code>/p/[slug]</code> .	

02. Persona Journeys & System Interaction Flows

Persona 1: The Candidate / Builder

ROLE: USER

Flow: Discovery → Challenge → Test → Verification → Dashboard

- Challenge Selection & Specification Intake (`/ideas/[id]`)**
Candidate selects a production problem (e.g., distributed deadlock detector), inspects architectural constraints, mock fleet topologies, and test criteria.
- Solution Submission & Gated Mutation (`SubmitSolutionModal`)**
Candidate submits Repo URL, Commit SHA, and PR Number. If unauthenticated, contextual `AuthModal` intercepts without losing user state.
- AI Scrutiny Audit & ATS Bullet Synthesis (`/jobs/[id]/apply`)**
Candidate triggers AI audit against job specifications. Streaming logs provide real-time feedback on AST inspection, latency metrics, and CV-ready impact bullets.
- Ledger Telemetry & Profile Management (`/dashboard` & `/settings`)**
Candidate monitors stamped proofs, 365-day certificate countdowns, and updates skill chips with optimistic UI rollback protection.

Persona 2: The Technical Recruiter / Hiring Manager

EXTERNAL VERIFIER

Flow: CV Link → Ledger Audit → Proof Inspection → ATS Export

- Public Portfolio Access (`/p/[slug]`)**
Zero login friction. Recruiter views the candidate's verified proof ledger. If viewer is unauthenticated, sensitive edit controls are hidden automatically.
- Cryptographic Proof & Telemetry Inspection**
Recruiter reviews SHA-256 commit stamps, p99 latency graphs, throughput figures, and mock fleet CI pass rates instead of unverified self-ratings.
- 1-Click ATS Docket Export**
Recruiter copies pre-formatted ATS-safe markdown summary to paste directly into Greenhouse, Lever, or Workday candidate evaluation forms.

- 1

Edge RBAC Inspection (proxy.ts)

Visiting /admin triggers Edge verification. Non-authorized users are rewritten to /403 without leaking admin UI assets.
- 2

Submission Verification & Stamping (/admin)

Auditor inspects submitted code diffs, verifies mock fleet test logs, and issues cryptographically signed 365-day certificates.
- 3

Production Problem Seeding

Admin publishes new architectural challenges with mock fleet definitions and automated evaluation harnesses into the live Idea Bank.

03. Edge Cases Diagnosed & Resolved in Code

AREA / COMPONENT	EDGE CASE IDENTIFIED	IMPLEMENTED SOLUTION
Reviewer RBAC Clearance AuthGuard.tsx & /admin	AuthGuard previously checked requireRole="admin" , blocking reviewer users even though Edge proxy allowed them.	Added allowedRoles={['admin', 'reviewer']} to AuthGuard , enabling both roles to access audit consoles seamlessly.
Unauthenticated Dashboard src/app/dashboard/page.tsx	Dashboard lacked client-side guard; client-side route transitions without cookies showed empty guest state.	Wrapped page in <AuthGuard fallbackMessage="..."> providing complete dual-layer defense.
Public Profile Ownership Leak src/app/p/[slug]/page.tsx	Unauthenticated visitors viewing /p/junior_dev saw "Edit Profile" because guest store defaulted to junior_dev .	Enforced isLoggedIn && slug === user.username to restrict edit controls strictly to authenticated owners.
Authenticated Login Loop src/app/login/page.tsx	Signed-in users navigating to /login were greeted by an orphaned login form.	Added reactive useEffect redirecting active sessions to callbackUrl or /dashboard immediately.
ESLint Vendor Script Warnings eslint.config.mjs	94 lint warnings fired across internal tool scripts in .agents/ .	Added ".agents/**" to globalIgnores , reducing project lint output to pristine 0 errors, 0 warnings.

04. Cross-Functional Team Contracts & Deliverables

The frontend is fully built, typed, and resilient with fallback adapters. To complete production rollout, here is exactly what the frontend requires from each engineering department:

Backend Engineering Team (Core Services)

TARGET: REST API / GO OR NODE

Implement endpoints conforming to the frontend TypeScript service layer contracts (frontend/src/services/):

Auth Endpoints:

POST /api/auth/github , POST /api/auth/email/magic-link , POST /api/auth/logout . Must return { token, role, expiresAt } and set HTTP-only cookies devledgr_session & devledgr_role .

User Profile CRUD:

GET /api/user/me , PUT /api/user/me . Must strictly accept and validate against Zod UserProfileUpdateSchema (headline, bio, statedSkills, plan, apiKey, githubUrl).

Idea Bank Service:

GET /api/ideas , GET /api/ideas/:id , POST /api/ideas (admin only). Must deliver mock fleet definitions and CI test specs.

Submissions & Stamping Engine:

POST /api/submissions , POST /api/submissions/:id/verify . Must trigger test harness, record p99 latency / throughput telemetry, and generate cryptographic SHA-256 certificate hashes valid for 365 days.

AI / LLM & Gateway Team

TARGET: AI GATEWAY / PYTHON OR FASTIFY

Provide high-throughput, low-latency endpoints matching the IAIService contract:

Scrutiny Audit Gateway:

POST /api/ai/audit . Accepts { repoUrl, commitHash, jobRequirements, submissions } . Streams progressive evaluation steps (AST analysis, concurrency verification, load test results) and returns final score (0-100) + tailored ATS markdown bullets.

Coaching Advice Service:

POST /api/ai/coach . Accepts { itineraryTitle, milestoneTitle, prompt } . Returns targeted architectural critiques and consensus debugging guidance.

Timeout & Resilience SLA:

Gateway must respond within REQUEST_TIMEOUT_MS (default: 15,000ms). If exceeded, frontend safely activates the mutation safety invariant.

DevOps, SRE & Cloud Infrastructure

TARGET: KUBERNETES / DOCKER / CDN

- Mock Fleet Topologies:** Deploy lightweight, containerized mock fleets (e.g. simulated Kafka brokers, Redis clusters, and HTTP endpoints) that candidate code can execute against during verification tests.
- Automated CI Sandbox Runner:** Secure execution environment (gVisor or Firecracker microVMs) to run candidate GitHub repositories against problem test harnesses in < 90 seconds.
- Environment Variables Provisioning:** Configure production variables across Edge runtime: `API_URL` , `AI_GATEWAY_URL` , `SESSION_SECRET` , `USE MOCKS=false` , `GITHUB_CLIENT_ID` , `GITHUB_CLIENT_SECRET` .
- Public Proof Subdomains:** Support wildcard DNS routing (e.g., `*.devledgr.io`) pointing to `/p/[slug]` for custom vanity ledger addresses.

Product, QA & Reviewer Staff

TARGET: QUALITY & AUDIT OPS

- Problem Seeding Roadmap:** Author initial batch of 15 authentic engineering challenges across Distributed Systems, FinTech, and Data Engineering with explicit failure topologies.
- Review SLA:** Establish 24-hour turnaround guideline for human-in-the-loop review tickets in the `/admin` verification cockpit.
- ATS Compatibility Testing:** Verify that exported recruiter markdown bullets correctly parse into major ATS platforms (Greenhouse, Lever, Ashby, Workday) without stripping formatting.

05. Essential API Contracts (Reference Schemas)

Submission Verification Payload

```
POST /api/submissions
Content-Type: application/json
Authorization: Bearer <session_token>

{
  "ideaId": "distributed-deadlock-detector",
  "repoUrl": "https://github.com/alex/deadlock-det",
  "commitHash": "a3f9d21b7c4e",
  "prNumber": 14,
  "architectureNotes": "Implemented Wait-For graph with cycle detection using Tarjan's algorithm."
}
```

AI Scrutiny Audit Response

```
POST /api/ai/audit (200 OK)
Content-Type: application/json

{
  "auditScore": 96,
  "verdict": "PASSED_STAMP_RECOMMENDED",
  "generatedBullets": [
    "Architected cycle-detection engine processing 10k req/s with",
    "Eliminated distributed deadlocks under simulated network p",
  ],
  "latencyMetrics": { "p99": "36ms", "throughput": "220 req/s" }
```

06. Implementation Timeline & Rollout Phases

PHASE	TARGET MILESTONE	KEY OUTPUT	RESPONSIBLE TEAM
Phase 1 (Completed)	Frontend Architecture & Service Layer	Next.js 16 app, adapter pattern, auth UI, profile editing, 0 lint warnings, 0 build errors.	Frontend Engineering
Phase 2 (Next 2 Weeks)	Live Backend & Auth Wiring	Deploy live auth & user services; flip <code>USE MOCKS=false</code> on staging; verify Edge cookies.	Backend + Frontend
Phase 3 (Weeks 3–4)	AI Gateway & Mock Fleet Runner	Live AST inspection endpoint, automated CI test runner with Dockerized microservice fleets.	AI Team + DevOps
Phase 4 (Week 5)	Public Launch & Recruiter Rollout	Seed 15 challenges, enable wildcard subdomains, open public signups with GitHub verification.	Product + All Teams

Sign-Off & Architecture Alignment: This document serves as the agreed technical baseline between Frontend, Backend, AI, and DevOps teams. Any changes to service interfaces, schemas, or cookie parameters require an ADR update in `frontend/docs/architecture/` .